Target Trial Emulation

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Project Leader / Real-World Operation

KSPM RWE SIG
October 23, 2025

Agenda

- Update: Externally Controlled Trial
- Target Trial Emulation
- Case Study

[참고]

✓ 김영도 박사님 발표 (2025-04, KSPM RWE SIG)

✓ 신주영 교수님 발표 (2025-09, KSPM MA Forum)

FDA: RWE in Regulatory Decision Making

- Approval, Safety related labeling changes
- As of **2025-09-26**



[Reference] https://www.fda.gov/science-research/real-world-evidence/fda-use-real-world-evidence-regulatory-decision-making

FDA: RWE in Regulatory Decision Making

- Table 1. CDER drug approvals that used real-world evidence → 8 entries
- Table 2. **CDER** safety related labeling changes, communications and publications that used real-word evidence → **26 entries**
- Table 3. CDER safety related labeling changes or other safety related actions that used real-world evidence- studies conducted by sponsors → 9 entries
- Table 4. CBER biologics approvals that used real-word evidence → 4 entries
- Table 5. CBER safety labeling changes that used real-world evidence- FDA and sponsor conducted studies → 10 entries

[Reference] https://www.fda.gov/science-research/real-world-evidence/fda-use-real-world-evidence-regulatory-decision-making

[Example] Lazertinib

- Using a real-world external control



www.nature.com/scientificreports

scientific reports

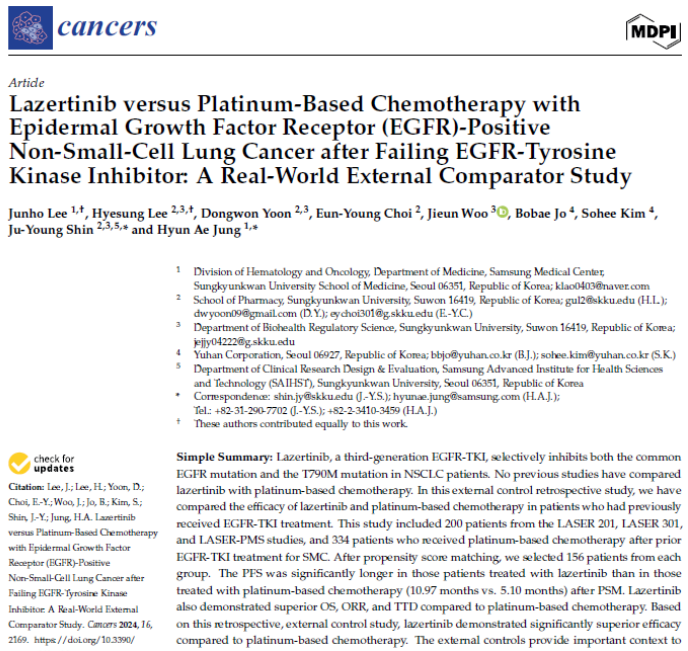
OPEN

Comparative effectiveness of lazertinib in patients with EGFR T790M-positive non-small-cell lung cancer using a real-world external control

Ha-Lim Jeon[1], Meesong Kwak[2], Sohee Kim[3], Hye-Yeon Yu[4], Ju-Young Shin[2,5,6]✉ & Hyun Ae Jung[1,6]✉

Lazertinib is a recently developed third-generation epidermal growth factor receptor (EGFR)-tyrosine kinase inhibitors used for patients with advanced *EGFR* T790M-positive non-small-cell lung cancer. We evaluated the effectiveness of lazertinib compared with osimertinib using an external control. We obtained individual patient data for the lazertinib arm from the LASER201 trial and the osimertinib arm from registry data at the Samsung Medical Center. In total, 75 and 110 patients were included in the lazertinib and osimertinib groups, respectively. After propensity score matching, each group had 60 patients and all baseline characteristics were balanced. The median follow-up duration was 22.0 and 29.6 months in the lazertinib and osimertinib group, respectively. The objective response rate (ORR) were 76.7% and 86.7% for lazertinib and osimertinib, respectively ($p = 0.08$). The median progression-free survival (PFS) was 12.3 months (95% confidence interval [CI] 9.5–19.1) and 14.4 months (95% CI 11.8–18.1) for the lazertinib and osimertinib group, respectively (hazard ratio [HR] 0.97; 95% CI 0.64–1.45, $p = 0.86$). The median overall survival with lazertinib was not reached and that with osimertinib was 29.8 months (HR 0.44; 95% CI 0.25–0.77, $p = 0.005$). Our study suggests that lazertinib has an ORR and PFS comparable to those of osimertinib and has the potential for superior survival benefits.

Keywords Lazertinib, Osimertinib, External control, Comparative effectiveness, *EGFR* T790M-positive non-small-cell lung cancer



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Article

Lazertinib versus Platinum-Based Chemotherapy with Epidermal Growth Factor Receptor (EGFR)-Positive Non-Small-Cell Lung Cancer after Failing EGFR-Tyrosine Kinase Inhibitor: A Real-World External Comparator Study

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† These authors contributed equally to this work.

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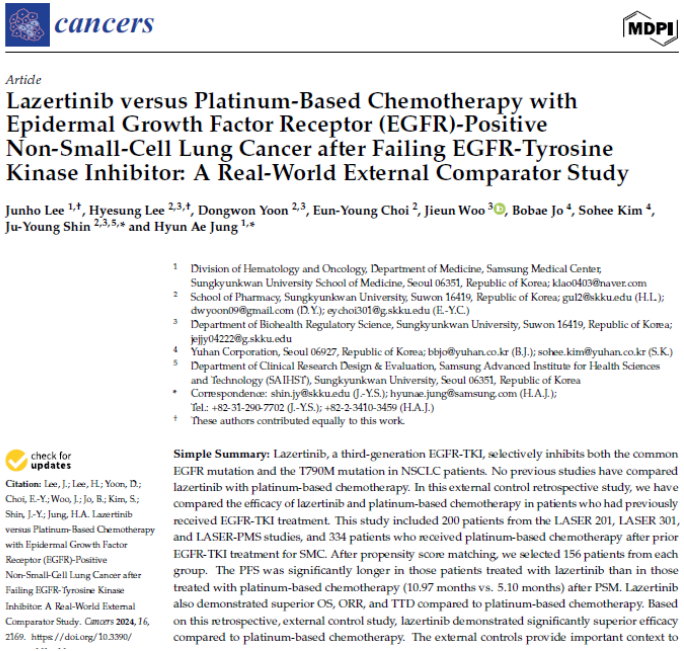
Lazertinib: Real-world External Control

[Data Sources] **Individual-patient-level data**

- **Lazertinib: LASER 201, LASER 301, and LASER PMS studies**
- External comparator group: Samsung Medical Center-Clinical Data Warehouse (**SMC-CDW**) = ROOT-HEALTH

[Study Population and Exposure]

- Diagnosis: Advanced NSCLC - between 1 June 2015 and 30 April 2022
- Lazertinib Treatment: Between 27 November 2017 and 30 April 2022
- The external control group: During the same time frame
  - **The index date**: **First date** of lazertinib treatment for the lazertinib group and the first date of platinum-based chemotherapy treatment for the external comparator group
  - All eligible patients scored 0 or 1 in ECOG performance status
  - We excluded patients who needed to be treated for or had a history of treatment for brain metastases within 14 days preceding the index date and patients who needed to be treated for or had a history of treatment for double primary cancer before the index date



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Article

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Lazertinib: Real-world External Control

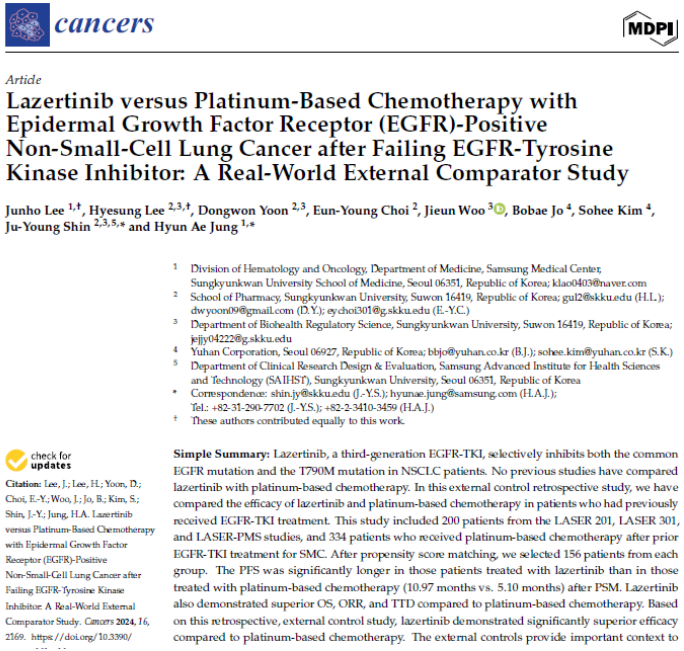
[Potential Confounders]

- Demographic characteristics: Age and sex
- Clinical characteristics: smoking status, ECOG performance status, tumor histology, American Joint Committee on Cancer Tumor-Node-Metastases stage, and brain metastasis
- The biomarker-related characteristics: EGFR mutation status
- Cancer-treatment-related characteristics: previous lines of systemic therapy, previous lines of EGFR-TKI, type of previous EGFR-TKI, time from immediate previous EGFR-TKI, and duration of previous EGFR-TKI

➔ The demographic characteristics were assessed at the **index date**, and other characteristics were assessed before the index date

[Statistical Analysis]

- 1:1 propensity score matching (**PSM**)
- Sensitivity analyses: Inverse probability of treatment weighting (**IPTW**) to address imbalances of potential confounders between the lazertinib and the external comparator groups



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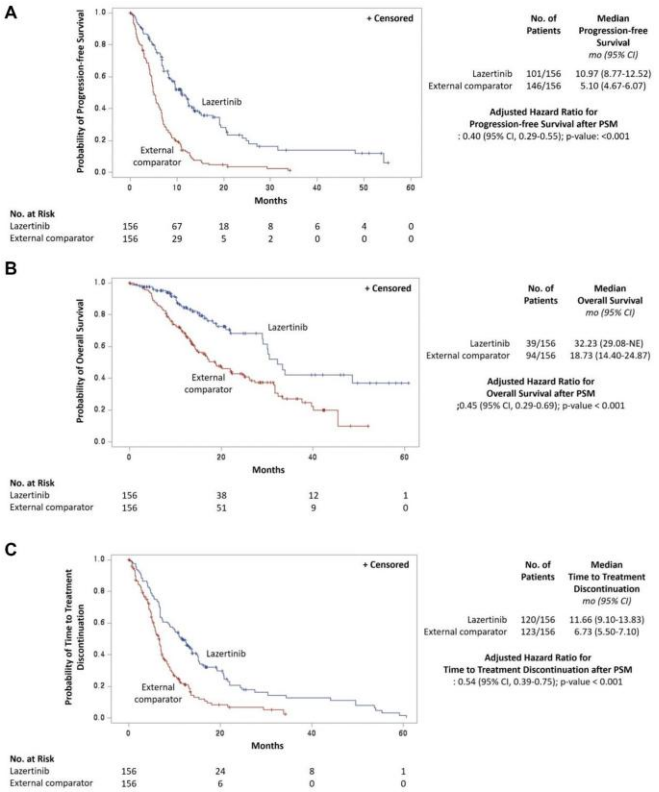
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Lazertinib: Real-world External Control

Table 1. Baseline characteristics.

Absolute standardized difference (aSD)
- significant imbalances between exposure groups

	Before PS Matching			After PS Matching		
	Lazertinib (*n* = 200)	External Comparator (*n* = 334)	aSD	Lazertinib (*n* = 156)	External Comparator (*n* = 156)	aSD
Age, median (Q1–Q3)	64 (56–71)	63 (56–70)	0.11	63.5 (55–70.5)	62.5 (57–69)	0.03
Female sex, *n* (%)	114 (57.0)	191 (57.2)	0.00	85 (54.5)	87 (55.8)	0.03
No history of smoking, *n* (%)	123 (61.5)	213 (63.8)	0.05	94 (60.3)	94 (60.3)	0.00
ECOG performance status, *n* (%)						
0	41 (20.5)	19 (5.7)	0.45	16 (10.3)	17 (10.9)	0.02
1	159 (79.5)	315 (94.3)		140 (89.7)	139 (89.1)	
Adenocarcinoma tumor histology, *n* (%)	194 (97.0)	320 (95.8)	0.06	151 (96.8)	150 (96.1)	0.03
Brain metastasis, *n* (%)	76 (38.0)	90 (27.0)	0.24	56 (35.9)	56 (35.9)	0.00
EGFR mutation status, *n* (%)						
L858R	37 (18.5)	145 (43.4)	0.56	27 (17.3)	67 (43.0)	0.58
Exon19Del	80 (40.0)	179 (53.6)	0.27	68 (43.6)	83 (53.2)	0.19
Others (G719X, L861Q, others, unknown)	83 (41.5)	19 (5.7)	0.35	61 (39.1)	8 (5.1)	0.90
Previous lines of systemic therapy, median (min–max)	1 (1–10)	1 (1–5)	0.45	1 (1–5)	1 (1–5)	0.06
Previous lines of EGFR-TKI treatment, median (min–max)	1 (1–10)	1 (1–5)	0.36	1 (1–3)	1 (1–5)	0.03
Type of previous EGFR-TKI treatment, median (min–max)						
Gefitinib	120 (60.0)	117 (35.0)	0.52	94 (60.3)	54 (34.6)	0.53
Erlotinib	25 (12.5)	77 (23.1)	0.28	15 (9.6)	43 (27.6)	0.47
Afatinib	61 (30.5)	193 (57.8)	0.57	46 (29.5)	89 (57.1)	0.58



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Target Trial Emulation (TTE)

- JAMA (2025-09-03)
  - Transparent Reporting of Observational Studies Emulating a Target Trial—The TARGET Statement
  - **Conclusions and Relevance:** Application of the TARGET guideline recommendations aims to improve reporting transparency and peer review and help researchers, clinicians, and other readers interpret and apply the results



Clinical Review & Education

JAMA | Special Communication

Transparent Reporting of Observational Studies Emulating a Target Trial—The TARGET Statement

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IMPORTANCE When randomized trials are unavailable or not feasible, observational studies can be used to answer causal questions about the comparative effects of interventions by attempting to emulate a hypothetical pragmatic randomized trial (target trial). Published guidance to aid reporting of these studies is not available.

OBJECTIVE To develop consensus-based guidance for reporting observational studies performed to estimate causal effects by explicitly emulating a target trial.

DESIGN, SETTING, AND PARTICIPANTS The Transparent Reporting of Observational Studies Emulating a Target Trial (TARGET) guideline was developed using the Enhancing the Quality and Transparency of Health Research (EQUATOR) framework. The development included (1) a systematic review of reporting practices in published studies that had explicitly aimed to emulate a target trial; (2) a 2-round online survey (August 2023 to March 2024; 18 expert participants from 6 countries) to assess the importance of candidate items selected from previous research and to identify additional items; (3) a 3-day expert consensus meeting (June 2024; 18 panelists) to refine the scope of the guideline and draft the checklist; and (4) pilot of the draft checklist with stakeholders (n = 108; September 2024 to February 2025). The checklist was further refined based on feedback on successive drafts.

FINDINGS The 21-item TARGET checklist is organized into 6 sections (abstract, introduction, methods, results, discussion, other information). TARGET provides guidance for reporting observational studies of interventions explicitly emulating a parallel group, individually randomized target trial, with adjustment for baseline confounders. Key recommendations are to (1) identify the study as an observational emulation of a target trial; (2) summarize the causal question and reason for emulating a target trial, (3) clearly specify the target trial protocol (ie, the causal estimand, identifying assumptions, data analysis plan) and how it was mapped to the observational data, and (4) report the estimate obtained for each causal estimand, its precision, and findings from additional analyses to assess the sensitivity of the estimates to assumptions, and design and analysis choices.

CONCLUSIONS AND RELEVANCE Application of the TARGET guideline recommendations aims to improve reporting transparency and peer review and help researchers, clinicians, and other readers interpret and apply the results.

Supplemental content

CME at jamacmelookup.com

[Reference] Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial-The TARGET Statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

The TARGET Statement

[Transparent Reporting of **Observational Studies** Emulating a Target Trial]

- **Importance**
  - When randomized trials are unavailable or not feasible, observational studies can be used **to answer causal questions** about the **comparative effects of interventions** by attempting to emulate a hypothetical pragmatic randomized trial (target trial)
  - Published guidance to aid reporting of these studies is not available
- **Objective**
  - To develop consensus-based guidance for reporting observational studies performed **to estimate causal effects by explicitly emulating a target trial**

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The TARGET Statement

- **Design, Setting, and Participants**
  - **Enhancing the Quality and Transparency of Health Research (EQUATOR) framework**

  - The development included

① **A systematic review** of reporting practices in published studies that had explicitly aimed to emulate a target trial

② A 2-round **online survey** (August 2023 to March 2024; 18 expert participants from 6 countries) to assess the importance of candidate items selected from previous research and to identify additional items

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④ A pilot of the draft **checklist** with stakeholders (n = 108; September 2024 to February 2025)
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The TARGET Statement

- **Findings**
  - The 21-item TARGET checklist is organized into 6 sections (**abstract, introduction, methods, results, discussion, other information**)
  - TARGET provides guidance for reporting observational studies of interventions explicitly emulating a parallel group, individually randomized target trial, with adjustment for baseline confounders
  - Key recommendations
    - ① **Identify** the study as an **observational emulation of a target trial**
    - ② **Summarize** the **causal question** and **reason for emulating a target trial**
    - ③ **Clearly specify** the **target trial protocol** (ie, the causal estimand, identifying assumptions, data analysis plan) and how it was **mapped to the observational data**
    - ④ **Report** the **estimate** obtained for each causal estimand, its precision, and findings from additional analyses to assess the **sensitivity of the estimates** to assumptions, and design and analysis choices

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Glossary of Terms Commonly Used

Immortal time
Time zero

Table 1. Glossary of Terms Commonly Used in Studies Emulating a Target Trial[a]

Term	Explanation
Causal contrast **ITT** **PP**	The comparison of the outcome distributions under 2 different treatment strategies. Examples of causal contrasts are being assigned to one vs another treatment strategy, irrespective of whether people actually receive the treatment assigned (the intention-to-treat effect), and the effect of fully adhering to one vs another treatment strategy as specified by the protocol (the per-protocol effect).
Causal estimand	The causal quantity (typically, an effect) that would be estimated to answer a causal question of interest. Key components of the causal estimand are the target population (eligibility criteria), the treatment strategies, the outcome, the timing of follow-up, and the causal contrast (see above).
Confounding	Confounding occurs when groups receiving different treatment strategies differ in their distribution of prognostic factors at time zero.[17] When there is confounding, differences in the outcome distribution between treatment groups may be explained by differences in prognostic factors rather than differences in treatment. A key concern for target trial emulation is whether some confounding bias remains after adjustment for measured confounders.
Design-related biases	Design-related biases refer to those that arise from decisions made by investigators when designing their analyses of observational data,[1,5] rather than biases associated with the observational study design (eg, confounding, measurement error). Common design-related biases include selection and misclassification, which may produce periods of time in the analysis when an individual cannot develop an outcome of interest (immortal time). These biases are due to misalignment of the start of follow-up (time zero) with the time an individual becomes eligible and is classified into a treatment strategy; selection bias may arise when eligibility occurs after assignment to a treatment strategy (selection depends on the outcome), and misclassification bias may arise when assignment occurs after eligibility (classification depends on the outcome).[6,18]
Identifying assumptions	Informally, assumptions that link the causal estimand to the observed data. Some examples of identifying assumptions for the per-protocol effect are[19] (1) Within levels of the adjustment variables, groups receiving each treatment strategy at each time have the same counterfactual risk of the outcome (conditional exchangeability). (2) For every combination of the adjustment variables, there is a nonzero (ie, positive) probability of receiving each treatment strategy at each time. (3) The treatment strategies are sufficiently well-defined.

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TTE: Glossary of Terms Commonly Used

- Target Trial Framework
  - ① **Specification of the target trial**
  - ② **Mapping that target trial to the data (emulation)**

Target trial framework	A methodological framework for causal inference from observational data, which applies the design principles of randomized trials. This involves designing observational analyses to explicitly emulate a hypothetical pragmatic randomized trial that would answer the question at hand: the target trial. The framework has 2 components: specification of the target trial and mapping that target trial to the data (emulation).

[a] Explanations adapted from Hernán et al 2020[19] and Hernán et al 2025.[1]

[Reference] Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial-The TARGET Statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

TARGET Checklist:
Abstract, Introduction, Methods

Table 2. TARGET Checklist of Recommended Items to Address in Reports of Studies Emulating a Target Trial[a]

Item No.	Checklist item	
Abstract		
1	a	Identify that the study attempts to emulate a target trial using observational data. State the study objectives and briefly summarize the specified target trial.
	b	Report the data sources used for emulation.
	c	Summarize key assumptions, statistical methods, findings and conclusions.
Introduction		
2	Background	Describe the scientific background of the study and the gap in knowledge.
3	Causal question	Summarize the causal question.
4	Rationale	Describe the rationale for emulating a target trial with the available data. Cite randomized trials informing the design of the target trial if applicable.
Methods		
5	Data sources	Cite the data sources contributing to the analyses and for each one describe the following: original purpose, type, the geographic locations, setting and time-period. If relevant, describe how the data were linked or pooled.

[Reference] Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial-The TARGET Statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

TARGET Checklist:
Target Trial Specification & Target Trial Emulation

- Eligibility criteria
- Treatment strategies
- Assignment procedures
- Follow-up
- Outcomes
- Casual contrasts
- **Identifying assumption**
- **Data analysis plan**

	Target trial specification		Target trial emulation	
6	Specify the components of the target trial protocol that would answer the causal question.	7	Describe how the components of the target trial protocol were emulated with the observational data, including how all variables were measured or ascertained.	
	Eligibility criteria		Eligibility criteria	
	a	Describe the eligibility criteria.	a	Describe how the eligibility criteria were operationalized with the data.
	Treatment strategies		Treatment strategies	
	b	Describe the treatment strategies that would be compared.	b	Describe how the treatment strategies were operationalized with the data.
	Assignment procedures		Assignment procedures	
	c	Report that eligible individuals would be randomly assigned to treatment strategies and may be aware of their treatment allocation.	c	Describe how assignment to treatment strategies was operationalized with the data.
	Follow-up		Follow-up	
	d	Clarify that follow-up would start at time of assignment to the treatment strategies. Specify when follow-up would end.	d	Clarify that follow-up starts at the time individuals were assigned to the treatment strategies. Describe how the end of follow-up was operationalized with the data.
	Outcomes		Outcomes	
	e	Describe the outcomes.	e	Describe how the outcomes were operationalized with the data.
	Causal contrasts		Causal contrasts	
	f	Describe the causal contrasts of interest, including effect measures.	f	Describe how the causal contrasts were operationalized with the data, including effect measures.
	Identifying assumptions		Identifying assumptions	
	g	Describe assumptions that would be made to identify each causal estimand. Describe the variables, if any, related to these assumptions.	g.i	For each causal estimand, describe assumptions made to identify it, including assumptions regarding baseline confounding due to lack of randomization.
			g.ii	Describe how the variables related to these assumptions were operationalized with the data.
	Data analysis plan		Data analysis plan	
	h	For each causal estimand, describe the data analysis procedures and any associated statistical modeling assumptions, including approaches for handling missing data.	h.i	For each causal estimand, describe the data analysis procedures and any associated statistical modeling assumptions, including approaches for handling missing data.
			h.ii	For each causal estimand, describe any additional analyses conducted to assess the sensitivity of the results to the choice of operationalizations, assumptions and analysis.

[Reference] Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial-The TARGET Statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

TARGET Checklist:
Results, Discussion & Other Information

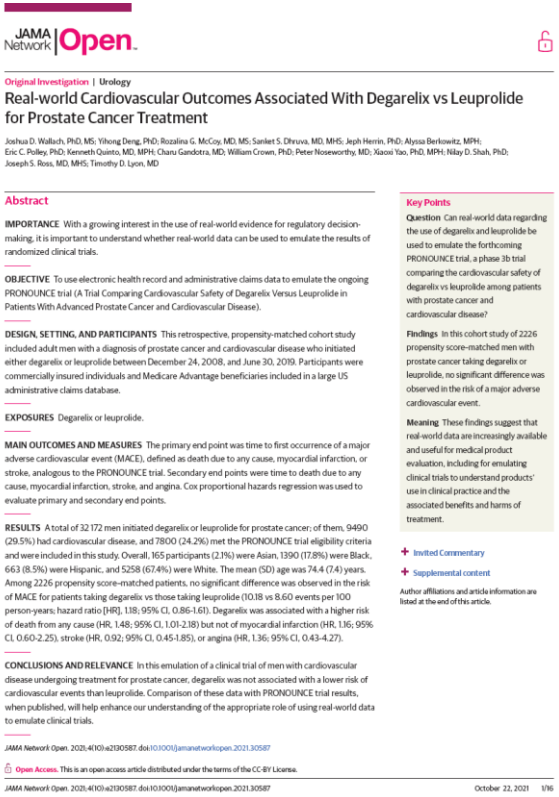
Results		
8	Participant selection	Report numbers of individuals assessed for eligibility, eligible, and assigned to each treatment strategy. A flow diagram is strongly recommended.
9	Baseline data	Describe the distribution of characteristics of individuals at baseline, by treatment strategy.
10	Follow-up	Summarize length of follow-up and describe reasons for end of follow-up for each treatment strategy and causal contrast.
11	Missing data	Describe the frequency of missing data in all variables, by treatment strategy when applicable.
12	Outcomes	Describe the frequency or distribution of each outcome, by treatment strategy.
13	Effect estimates	Report the effect estimates for each causal contrast with corresponding measures of precision, including both absolute and relative measures of effect, when applicable.
14	Additional analyses	Report results of all analyses to assess the sensitivity of the estimates to choices in operationalizations, assumptions and analysis.
Discussion		
15	Interpretation	Provide an interpretation of the key findings.
16	Limitations	Discuss the limitations of the study considering differences between the target trial and its emulation and the plausibility of assumptions, including assumptions regarding baseline confounding due to lack of randomization.
Other information		
17	Ethics	Provide the institutional research board or ethics committee that approved the study and approval numbers, if relevant.
18	Registration	State whether, when and where the study protocol was registered.
19	Sharing of study materials	Provide information on whether data, analytic code and/or other materials are accessible, and where and how they can be accessed.
20	Funding sources	Provide the sources of funding and detail the role of the funders in the design, conduct and reporting of the study.
21	Conflicts of interest	State any conflicts of interest and financial disclosures for all authors.

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[Reference] Cashin AG, Hansford HJ, Hernán MA, et al. Transparent Reporting of Observational Studies Emulating a Target Trial-The TARGET Statement. *JAMA*. 2025;334(12):1084-1093. doi:10.1001/jama.2025.13350

Case Study

- Real-world **Cardiovascular Outcomes** Associated With Degarelix vs Leuprolide for **Prostate Cancer Treatment**
  - JAMA Network | Open (2021-10-22)



JAMA Network | Open

Original Investigation | Urology

Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment

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Abstract

IMPORTANCE With a growing interest in the use of real-world evidence for regulatory decision-making, it is important to understand whether real-world data can be used to emulate the results of randomized clinical trials.

OBJECTIVE To use electronic health record and administrative claims data to emulate the ongoing PRONOUNCE trial (A Trial Comparing Cardiovascular Safety of Degarelix Versus Leuprolide in Patients With Advanced Prostate Cancer and Cardiovascular Disease).

DESIGN, SETTING, AND PARTICIPANTS This retrospective, propensity-matched cohort study included adult men with a diagnosis of prostate cancer and cardiovascular disease who initiated either degarelix or leuprolide between December 24, 2008, and June 30, 2019. Participants were commercially insured individuals and Medicare Advantage beneficiaries included in a large US administrative claims database.

EXPOSURES Degarelix or leuprolide.

MAIN OUTCOMES AND MEASURES The primary end point was time to first occurrence of a major adverse cardiovascular event (MACE), defined as death due to any cause, myocardial infarction, or stroke, analogous to the PRONOUNCE trial. Secondary end points were time to death due to any cause, myocardial infarction, stroke, and angina. Cox proportional hazards regression was used to evaluate primary and secondary end points.

RESULTS A total of 32 172 men initiated degarelix or leuprolide for prostate cancer; of them, 9490 (29.5%) had cardiovascular disease, and 7800 (24.2%) met the PRONOUNCE trial eligibility criteria and were included in this study. Overall, 165 participants (2.1%) were Asian, 1390 (17.8%) were Black, 663 (8.5%) were Hispanic, and 5258 (67.4%) were White. The mean (SD) age was 74.4 (7.4) years. Among 2226 propensity score–matched patients, no significant difference was observed in the risk of MACE for patients taking degarelix vs those taking leuprolide (10.18 vs 8.60 events per 100 person-years; hazard ratio [HR], 1.18; 95% CI, 0.86-1.61). Degarelix was associated with a higher risk of death from any cause (HR, 1.48; 95% CI, 1.01-2.18) but not of myocardial infarction (HR, 1.16; 95% CI, 0.60-2.25), stroke (HR, 0.92; 95% CI, 0.45-1.85), or angina (HR, 1.36; 95% CI, 0.43-4.27).

CONCLUSIONS AND RELEVANCE In this emulation of a clinical trial of men with cardiovascular disease undergoing treatment for prostate cancer, degarelix was not associated with a lower risk of cardiovascular events than leuprolide. Comparison of these data with PRONOUNCE trial results, when published, will help enhance our understanding of the appropriate role of using real-world data to emulate clinical trials.

Key Points

Question Can real-world data regarding the use of degarelix and leuprolide be used to emulate the forthcoming PRONOUNCE trial, a phase 3b trial comparing the cardiovascular safety of degarelix vs leuprolide among patients with prostate cancer and cardiovascular disease?

Findings In this cohort study of 2226 propensity score–matched men with prostate cancer taking degarelix or leuprolide, no significant difference was observed in the risk of a major adverse cardiovascular event.

Meaning These findings suggest that real-world data are increasingly available and useful for medical product evaluation, including for emulating clinical trials to understand products' use in clinical practice and the associated benefits and harms of treatment.

+ Invited Commentary
+ Supplemental content

Author affiliations and article information are listed at the end of this article.

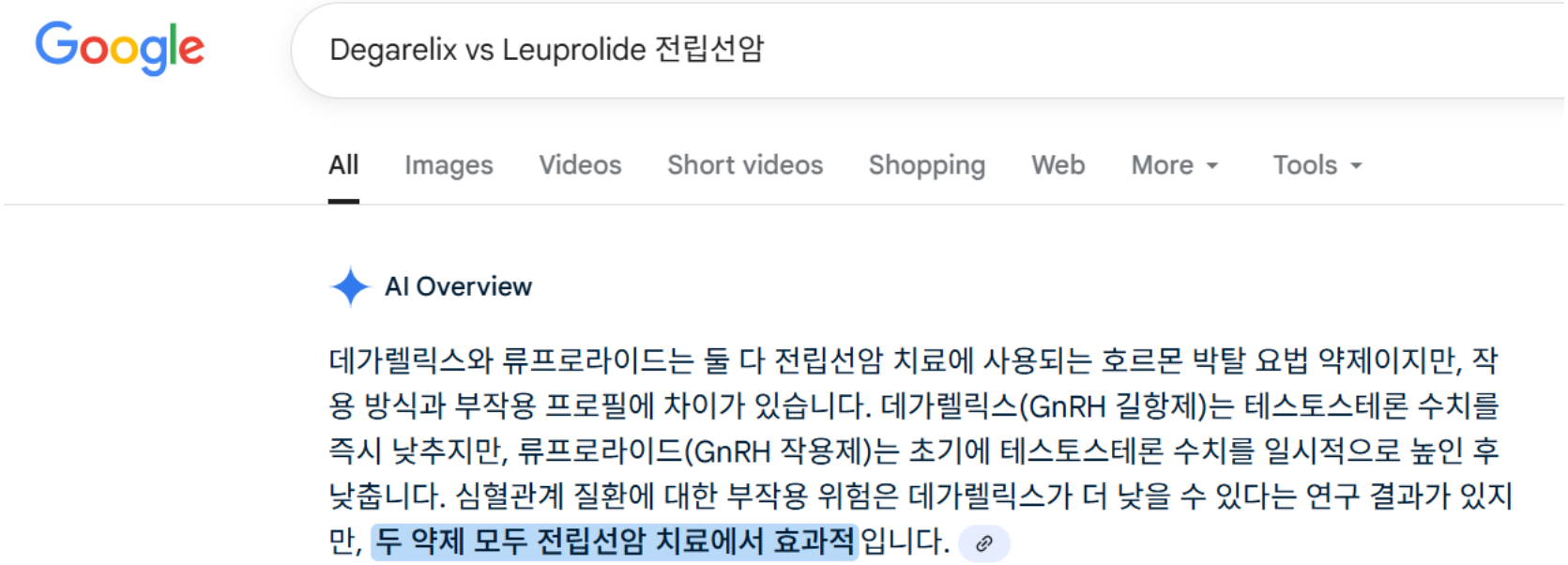
JAMA Network Open. 2021;4(10):e2130587. doi:10.1001/jamanetworkopen.2021.30587

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[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

(참고) 구글 검색: AI Overview

- Degarelix vs Leuprolide 전립선암



[Reference] google.com

(참고) 구글 검색: AI Overview

Degarelix (GnRH antagonist)



데가렐릭스 vs 류프로라이드 비교

구분	데가렐릭스 (Degarelix)	류프로라이드 (Leuprolide)
작용 방식	GnRH 길항제	GnRH 작용제
초기 테스토스테론 수치	즉시 감소	일시적으로 증가 후 감소
심혈관계 부작용 위험	낮은 경향	높은 경향 (연구에 따라 차이 있음)
효과	빠른 테스토스테론 억제	장기간 효과

결론

두 약제 모두 전립선암 치료에 효과적이지만, 치료 목표와 환자의 건강 상태에 따라 적합한 약제가 달라질 수 있습니다. 심혈관계 질환 위험이 높은 환자에게는 데가렐릭스가 더 나은 선택일 수 있으며, 빠른 호르몬 억제가 필요한 경우에는 데가렐릭스를 우선 고려할 수 있습니다. 의사는 환자의 상태와 치료 목표를 고려하여 가장 적합한 약제를 선택해야 합니다.

Dive deeper in AI Mode

This is for informational purposes only. For medical advice or a diagnosis, consult a professional. AI responses may include mistakes. Learn more

Leuprolide (GnRH agonist)



[Reference]
1. https://firmagon.com/hcp/dosing-and-administration/, 2. https://metrodrugs.com/patient-services/medication-information/leuprolide-acetate-2-week-kit-injection/

TTE: Degarelix vs Leuprolide - Cardiovascular Outcomes in patients with Prostate Cancer

[Key Points]

- **Question**
  - Can real-world data regarding the use of degarelix and leuprolide be used to emulate the forthcoming PRONOUNCE trial, a phase 3b trial comparing the cardiovascular safety of degarelix vs leuprolide among patients with prostate cancer and cardiovascular disease?
- **Findings**
  - In this cohort study of 2,226 propensity score–matched men with prostate cancer taking degarelix or leuprolide, **no significant difference was observed in the risk of a major adverse cardiovascular event**
- **Meaning**
  - These findings suggest that real-world data are increasingly available and useful for medical product evaluation, including for emulating clinical trials to understand products' use in clinical practice and the associated benefits and harms of treatment

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;*4*(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Degarelix vs Leuprolide - Abstract

- **Importance**
  - With a growing interest in the use of real-world evidence for regulatory decision-making, it is important to understand whether real-world data can be used to emulate the results of randomized clinical trials
- **Objective**
  - To use electronic health record and administrative claims data to emulate the ongoing PRONOUNCE trial (A Trial Comparing Cardiovascular Safety of Degarelix Versus Leuprolide in Patients With Advanced Prostate Cancer and Cardiovascular Disease)
- **Design, Setting, and Participants**
  - This **retrospective, propensity-matched cohort study** included adult men with a diagnosis of prostate cancer and cardiovascular disease who initiated either degarelix or leuprolide between **December 24, 2008, and June 30, 2019**
  - Participants were commercially insured individuals and Medicare Advantage beneficiaries included in a large US administrative claims database

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Degarelix vs Leuprolide - Abstract

- **Exposures:** Degarelix or leuprolide

- **Main Outcomes and Measures**
  - The primary end point: **time to first occurrence** of a **major adverse cardiovascular event (MACE)**, defined as death due to any cause, myocardial infarction, or stroke, analogous to the PRONOUNCE trial
  - Secondary end points: **time to death** due to any cause, myocardial infarction, stroke, and angina

→ Cox proportional hazards regression was used to evaluate primary and secondary end points

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;*4*(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Degarelix vs Leuprolide - Abstract

- **Results**
  - A total of 32,172 men initiated degarelix or leuprolide for prostate cancer → 9,490 (29.5%) had cardiovascular disease → 7,800 (24.2%) met the PRONOUNCE trial eligibility criteria and were included in this study
  - Overall, 165 participants (2.1%) were Asian, 1,390 (17.8%) were Black, 663 (8.5%) were Hispanic, and 5,258 (67.4%) were White / The mean (SD) age was 74.4 (7.4) years
  - Among 2,226 propensity score–matched patients, **no significant difference was observed in the risk of MACE** for patients taking degarelix vs those taking leuprolide (10.18 vs 8.60 events per 100 person-years; hazard ratio [HR], 1.18; 95% CI, 0.86-1.61)
  - **Degarelix was associated with a higher risk of death from any cause** (HR, 1.48; 95% CI, 1.01-2.18) but not of myocardial infarction (HR, 1.16; 95% CI, 0.60-2.25), stroke (HR, 0.92; 95% CI, 0.45-1.85), or angina (HR, 1.36; 95% CI, 0.43-4.27)

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Degarelix vs Leuprolide - Abstract

- **Conclusions and Relevance**
  - In this emulation of a clinical trial of men with cardiovascular disease undergoing treatment for prostate cancer, **degarelix was not associated with a lower risk of cardiovascular events than leuprolide**
  - Comparison of these data with PRONOUNCE trial results, when published, will help enhance our understanding of the appropriate role of using real-world data to emulate clinical trials

- **Supplement:** eAppendix. **Study Protocol**

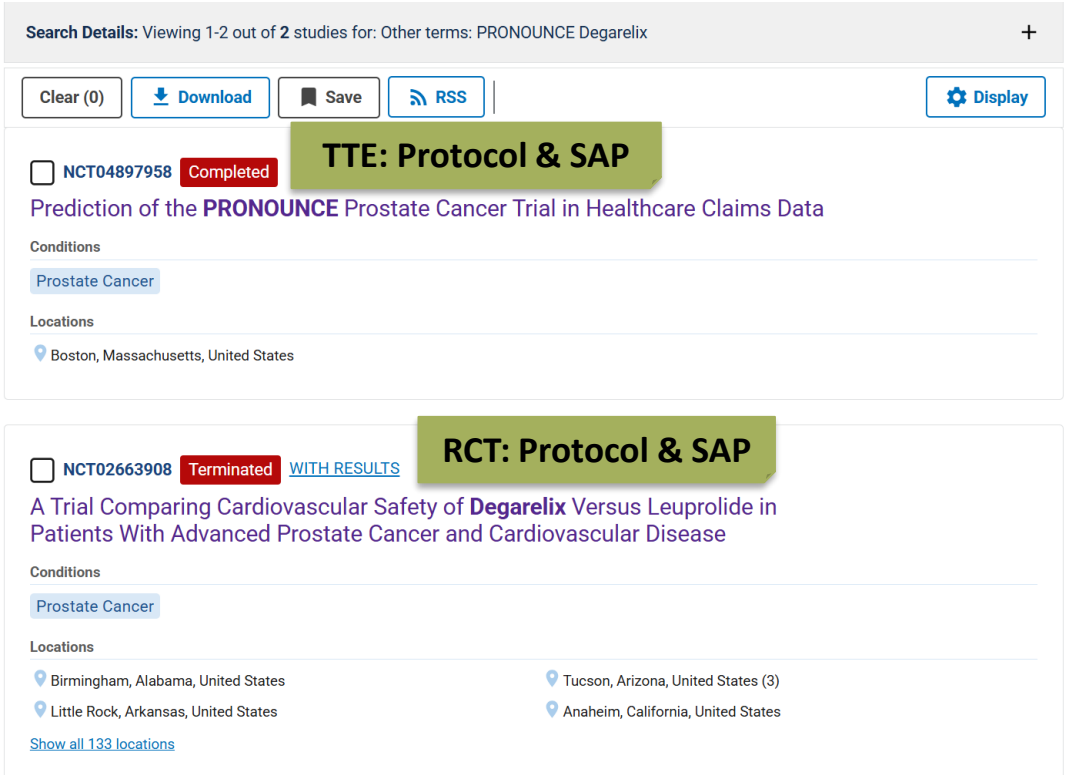
2. KEY DEFINITIONS

Item	Variable name	Description
Index Date	index_date	The date patients received degarelix (Firmagon) or leuprolide (Lupron depot)
Index Medication	index_med	Patient's first prescription of degarelix (Firmagon) or leuprolide (Lupron depot)
Baseline Period	baseline	Any time before and including the index date used to establish a patient's medical history
Study Period		12/24/2008 (degarelix (Firmagon) FDA approval date) – 6/30/2019
Data Source		OptumLabs Data Warehouse

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

Randomized Controlled Trial (RCT)

- A Trial Comparing Cardiovascular Safety of Degarelix Versus Leuprolide in Patients With Advanced Prostate Cancer and Cardiovascular Disease (PRONOUNCE)
  - ClinicalTrials.gov ID: NCT02663908
  - Sponsor: Ferring Pharmaceuticals
  - Study Start: 2016-04-19
  - Study Completion: 2021-03-29
  - **Enrollment (Actual): 545**



[Reference] https://clinicaltrials.gov/search?term=PRONOUNCE%20Degarelix

Randomized Controlled Trial (RCT)

Outcome Measures

Expand all / Collapse all

1. Time From Randomization to the First Confirmed (Adjudicated) Occurrence of the Composite Major Adverse Cardiovascular Event (MACE) Endpoint; Percentage of Observed Subjects With Outcome Measure Events During the Trial

Type: Primary | Time Frame: Randomization to Day 336 (end-of-trial)

Description	Composite MACE endpoint was defined as: death due to any cause, non-fatal myocardial infarction or non-fatal stroke. Kaplan Meier plots of the cumulative incidence rate by treatment groups were used to depict first confirmed (adjudicated) occurrence of composite MACE over time. Percentage of observed subjects with outcome measure events during the trial are reported. Subjects were censored at the time a subject discontinued the trial, was lost to follow-up, discontinued treatment with IMP, initiated treatment with prohibited medication (including hormonal combination therapy), or at Day 336, whichever occurred first.	
Time Frame	Randomization to Day 336 (end-of-trial)	
Analysis Population Description	The FAS consisted of all randomized and treated subjects (who received at least one dose of IMP) and was analyzed based on the planned (randomized) treatment.	
Arm/Group Title	Degarelix 240 mg/80 mg	Leuprolide 22.5 mg
Arm/Group Description	Degarelix 240 mg/80 mg: Degarelix at a starting dose of 240 mg administered as two SC depot injections, each containing 120 mg of degarelix; followed by up to 11 maintenance doses of 80 mg degarelix administered as single SC depot injections at monthly (28-day) intervals.	Leuprolide 22.5 mg: Leuprolide at dose of 22.5 mg administered as intramuscular depot injection every 3 months throughout the trial.
Overall Number of Participants Analyzed	275	269
Measure Type: Number | Unit of Measure: percentage of subjects	5.5	4.1

Statistical Analysis 1

Statistical Analysis Overview	
Comparison Group Selection	Degarelix 240 mg/80 mg, Leuprolide 22.5 mg
Comments	[Not Specified]
Type of Statistical Test	Other
Comments	[Not Specified]
Statistical Test of Hypothesis	
P-Value	0.5294
Comments	[Not Specified]
Method	Log Rank
Comments	The p-value of the log-rank test is based on comparison of the treatment groups stratified for age group and region.
Method of Estimation	
Estimation Parameter	Hazard Ratio (HR)
Estimated Value	1.283
Confidence Interval	(2-Sided) 95% 0.589 to 2.794
Estimation Comments	Baseline hazards for the Cox regression were stratified over variables: age group and region.

[Reference] https://www.clinicaltrials.gov/study/NCT02663908?tab=results

Target Trial Emulation

- **Prediction of the PRONOUNCE Prostate Cancer Trial in Healthcare Claims Data**
  - ClinicalTrials.gov ID: NCT04897958 / Sponsor: Brigham and Women's Hospital
- **Detailed Description**
  - A non-randomized, non-interventional study that is part of the **RCT DUPLICATE initiative** (www.rctduplicate.org) of the Brigham and Women's Hospital, Harvard Medical School
  - It is intended to replicate, as closely as is possible in healthcare insurance claims data, the trial
  - Although many features of the trial cannot be directly replicated in healthcare claims, key design features, including **outcomes, exposures, and inclusion/exclusion criteria**, were selected to proxy those features from the trial
  - Randomization is also not replicable in healthcare claims data but was proxied through a statistical balancing of measured covariates according to standard practice
  - Investigators assume that the **RCT provides the reference standard treatment effect estimate** and that failure to replicate RCT findings is indicative of the inadequacy of the healthcare claims data for replication for a range of possible reasons and does not provide information on the validity of the original RCT finding.

[Reference]
https://clinicaltrials.gov/study/NCT04897958?term=PRONOUNCE%20Degarelix&rank=1

TTE

- Data Source
  - 논문: OptumLabs Data Warehouse
- Design Diagram

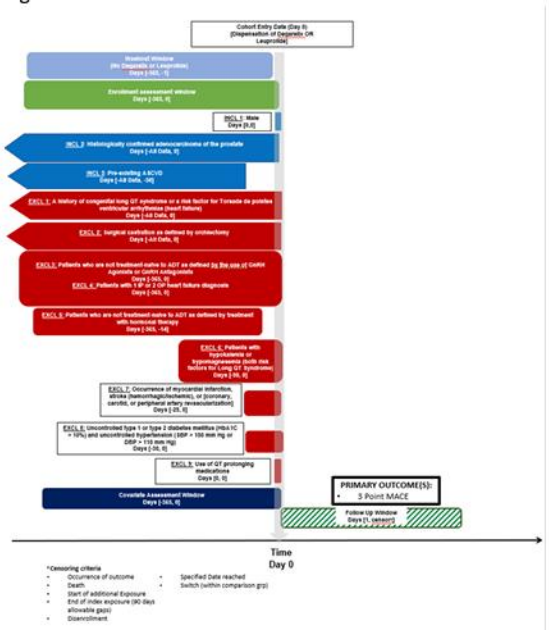
3. Data Source(s)

Optum CDM, IBM® MarketScan®, Medicare Diabetes

4. Study Design Diagram

The study design diagram visualizes key aspects of the longitudinal study design for expedited review.

Figure 1.



[Reference] https://clinicaltrials.gov/study/NCT04897958?term=PRONOUNCE%20Degarelix&rank=1

TTE: Baseline

Table 1. Baseline Characteristics Before and After PS Matching

Table 1. Baseline Characteristics Before and After PS Matching

Characteristic	Before PS matching				After PS matching			
	Patients, No. (%)				Patients, No. (%)			
	Degarelix (n = 1120)	Leuprolide (n = 6680)	Total (N = 7800)	SMD	Degarelix (n = 1113)	Leuprolide (n = 1113)	Total (N = 2226)	SMD
Age, y								
Mean (SD)	75.0 (7.6)	74.3 (7.4)	74.4 (7.4)	0.10	75.1 (7.6)	75.0 (7.4)	75.0 (7.5)	0.01
Median (IQR)	75.5 (70.0-81.0)	75.0 (70.0-80.0)	75.0 (70.0-80.0)		76.0 (70.0-81.0)	76.0 (71.0-81.0)	76.0 (70.0-81.0)	
≤54	<11[a]	55 (0.8)	>60[a]	0.04	<11[a]	11 (1.0)	>14[a]	0.05
55-64	113 (10.1)	662 (9.9)	775 (9.9)	0.01	111 (10.0)	96 (8.6)	207 (9.3)	0.05
65-74	>383[a]	2418 (36.2)	>2800[a]	0.03	>380[a]	367 (33.0)	>750[a]	0.03
≥75	613 (54.7)	3545 (53.1)	4158 (53.3)	0.03	611 (54.9)	639 (57.4)	1250 (56.2)	0.05
Race and ethnicity								
Asian	24 (2.1)	141 (2.1)	165 (2.1)	0.00	24 (2.2)	27 (2.4)	51 (2.3)	0.02
Black	209 (18.7)	1181 (17.7)	1390 (17.8)	0.03	207 (18.6)	208 (18.7)	415 (18.6)	0.00
Hispanic	55 (4.9)	608 (9.1)	663 (8.5)	0.17	55 (4.9)	51 (4.6)	106 (4.8)	0.02
White	783 (69.9)	4475 (67.0)	5258 (67.4)	0.06	778 (69.9)	779 (70.0)	1557 (69.9)	0.00
Unknown	49 (4.4)	275 (4.1)	324 (4.2)	0.01	49 (4.4)	48 (4.3)	97 (4.4)	0.00
Geographic region								
Midwest	301 (26.9)	1830 (27.4)	2131 (27.3)	0.01	301 (27.0)	313 (28.1)	614 (27.6)	0.02
Northeast	217 (19.4)	1063 (15.9)	1280 (16.4)	0.09	214 (19.2)	205 (18.4)	419 (18.8)	0.02
South	>495[a]	>3158[a]	>3660	0.05	>493[a]	>489[a]	>993[a]	0.01
West	96 (8.6)	618 (9.3)	714 (9.2)	0.02	94 (8.4)	95 (8.5)	189 (8.5)	0.00
Unknown	<11[a]	<11[a]	<11[a]	0.01	<11[a]	<11[a]	<11[a]	0.00
Serum PSA level[b]								
No. (%)	440 (35.2)	2246 (33.6)	2686 (34.4)	NA	423 (38.0)	423 (38.0)	846 (38.0)	NA
Mean (SD), ng/mL	74.3 (261.2)	48.9 (290.7)	53.0 (286.1)	0.09	70.6 (262.9)	59.3 (341.6)	64.9 (304.7)	0.05
Median (IQR), ng/mL	11.6 (6.2-32.4)	9.2 (5.1-19.8)	9.4 (5.2-21.6)		11.6 (6.2-30.8)	9.9 (5.3-24.4)	11.0 (5.8-27.3)	
eGFR[c]								
No. (%)	433 (34.6)	2353 (35.2)	2786 (35.7)	NA	428 (38.5)	377 (33.9)	805 (36.2)	NA
Mean (SD), mL/min/1.73 m^2	66.5 (20.5)	66.8 (20.4)	66.8 (20.4)	0.02	66.5 (20.5)	64.9 (21.4)	65.7 (20.9)	0.08
Median (IQR), mL/min/1.73 m^2	68.3 (52.4-81.9)	68.5 (53.1-82.5)	68.5 (53.0-82.5)		68.1 (52.4-81.9)	67.2 (51.6-80.4)	67.8 (52.1-81.2)	
Prostate biopsy within 6 mos of index date, No. (%)	682 (60.9)	3635 (54.4)	4317 (55.3)	0.13	676 (60.7)	683 (61.4)	1359 (61.1)	0.01
Baseline comorbidities								
Coronary artery disease	593 (52.9)	3424 (51.3)	4017 (51.5)	0.03	593 (53.3)	580 (52.1)	1173 (52.7)	0.02
Chronic kidney disease	167 (14.9)	877 (13.1)	1044 (13.4)	0.05	166 (14.9)	176 (15.8)	342 (15.4)	0.03
Congestive heart failure	191 (17.1)	1041 (15.6)	1232 (15.8)	0.04	190 (17.1)	173 (15.5)	363 (16.3)	0.04
Cerebrovascular disease	159 (14.2)	885 (13.2)	1044 (13.4)	0.03	158 (14.2)	169 (15.2)	327 (14.7)	0.03
Peripheral vascular disease	326 (29.1)	2001 (30.0)	2327 (29.8)	0.02	325 (29.2)	322 (28.9)	647 (29.1)	0.01
Obesity[d]	71 (6.3)	451 (6.8)	522 (6.7)	0.02	70 (6.3)	62 (5.6)	132 (5.9)	0.03
Atrial fibrillation	191 (17.1)	1104 (16.5)	1295 (16.6)	0.01	189 (17.0)	197 (17.7)	386 (17.3)	0.02
Sleep apnea	107 (9.6)	682 (10.2)	789 (10.1)	0.02	105 (9.4)	107 (9.6)	212 (9.5)	0.01
Hypertension	882 (78.8)	5290 (79.2)	6172 (79.1)	0.01	876 (78.7)	867 (77.9)	1743 (78.3)	0.02
MI	119 (10.6)	696 (10.4)	815 (10.4)	0.01	119 (10.7)	118 (10.6)	237 (10.6)	0.00
Stroke	60 (5.4)	375 (5.6)	435 (5.6)	0.01	59 (5.3)	68 (6.1)	127 (5.7)	0.04
PCI	81 (7.2)	494 (7.4)	575 (7.4)	0.01	81 (7.3)	83 (7.5)	164 (7.4)	0.01
CABG	97 (8.7)	634 (9.5)	731 (9.4)	0.03	97 (8.7)	90 (8.1)	187 (8.4)	0.02
PAD	232 (20.7)	1339 (20.0)	1571 (20.1)	0.02	229 (20.6)	228 (20.5)	457 (20.5)	0.00
Dementia	57 (5.1)	271 (4.1)	328 (4.2)	0.05	55 (4.9)	47 (4.2)	102 (4.6)	0.03
COPD	244 (21.8)	1340 (20.1)	1584 (20.3)	0.04	241 (21.7)	234 (21.0)	475 (21.3)	0.02
Peptic ulcer disease	11 (1.0)	73 (1.1)	84 (1.1)	0.01	11 (1.0)	16 (1.4)	27 (1.2)	0.04
Mild liver disease	78 (7.0)	524 (7.8)	602 (7.7)	0.03	78 (7.0)	77 (6.9)	155 (7.0)	0.00
Diabetes without chronic complication	414 (37.0)	2405 (36.0)	2819 (36.1)	0.02	411 (36.9)	375 (33.7)	786 (35.3)	0.07
Diabetes with chronic complication	182 (16.3)	1011 (15.1)	1193 (15.3)	0.03	180 (16.2)	184 (16.5)	364 (16.4)	0.01
Metastatic solid tumor	251 (22.4)	1097 (16.4)	1348 (17.3)	0.15	247 (22.2)	250 (22.5)	497 (22.3)	0.01
Rheumatic disease	21 (1.9)	149 (2.2)	170 (2.2)	0.03	21 (1.9)	28 (2.5)	49 (2.2)	0.04
Charlson comorbidity score								
Mean (SD)	5.5 (3.2)	5.1 (3.0)	5.1 (3.1)	0.14	5.5 (3.2)	5.5 (3.3)	5.5 (3.3)	0.00
Median (IQR)	5.0 (3.0-8.0)	4.0 (3.0-7.0)	4.0 (3.0-7.0)		5.0 (3.0-8.0)	4.0 (3.0-8.0)	5.0 (3.0-8.0)	
Radiotherapy within 6 mos before index date, No. (%)	19 (1.7)	204 (3.1)	223 (2.9)	0.09	19 (1.7)	17 (1.5)	36 (1.6)	0.01
Use of bicalutamide within 6 mos before index date, No. (%)	0	454 (40.8)	454 (40.8)	NA	NA	NA	NA	NA

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Baseline

Other baseline medications within 6 mos before index date								
Statin	717 (64.0)	4264 (63.8)	4981 (63.9)	0.00	714 (64.2)	723 (65.0)	1437 (64.6)	0.02
Nonstatin lipid-lowering medications	170 (15.2)	976 (14.6)	1146 (14.7)	0.02	169 (15.2)	173 (15.5)	342 (15.4)	0.01
ACEi	396 (35.4)	2458 (36.8)	2854 (36.6)	0.03	391 (35.1)	393 (35.3)	784 (35.2)	0.00
ARB	215 (19.2)	1243 (18.6)	1458 (18.7)	0.02	211 (19.0)	218 (19.6)	429 (19.3)	0.02
ACEi and ARB	593 (52.9)	3620 (54.2)	4213 (54.0)	0.03	588 (52.8)	591 (53.1)	1179 (53.0)	0.01
Sacubitril and valsartan	<11[a]	>15[a]	22 (0.3)	0.00	<11[a]	<11[a]	<11[a]	0.02
Warfarin	102 (9.1)	648 (9.7)	750 (9.6)	0.02	102 (9.2)	96 (8.6)	198 (8.9)	0.02
DOAC	70 (6.3)	298 (4.5)	368 (4.7)	0.08	68 (6.1)	70 (6.3)	138 (6.2)	0.01
β-blockers	581 (51.9)	3211 (48.1)	3792 (48.6)	0.08	577 (51.8)	577 (51.8)	1154 (51.8)	0.00
Loop diuretics	174 (15.5)	1042 (15.6)	1216 (15.6)	0.00	173 (15.5)	160 (14.4)	333 (15.0)	0.03
Aldosterone antagonist	46 (4.1)	230 (3.4)	276 (3.5)	0.04	45 (4.0)	52 (4.7)	97 (4.4)	0.03
Digoxin	28 (2.5)	205 (3.1)	233 (3.0)	0.04	28 (2.5)	30 (2.7)	58 (2.6)	0.01
Calcium channel blocker	294 (26.3)	1666 (24.9)	1960 (25.1)	0.03	291 (26.1)	269 (24.2)	560 (25.2)	0.05
Antiplatelet	214 (19.1)	1219 (18.2)	1433 (18.4)	0.02	211 (19.0)	214 (19.2)	425 (19.1)	0.01
No. of hospitalizations								
0	922 (82.3)	5529 (82.8)	6451 (82.7)	0.01	915 (82.2)	894 (80.3)	1809 (81.3)	0.05
1	153 (13.7)	915 (13.7)	1068 (13.7)	0.00	153 (13.7)	176 (15.8)	329 (14.8)	0.06
≥2	45 (4.0)	236 (3.5)	281 (3.6)	0.03	45 (4.0)	43 (3.9)	88 (4.0)	0.01
No. of ED visits								
0	837 (74.7)	5151 (77.1)	5988 (76.8)	0.06	835 (75.0)	834 (74.9)	1669 (75.0)	0.00
1	164 (14.6)	935 (14.0)	1099 (14.1)	0.02	160 (14.4)	155 (13.9)	315 (14.2)	0.01
≥2	119 (10.6)	594 (8.9)	713 (9.1)	0.06	118 (10.6)	124 (11.1)	242 (10.9)	0.02
Year of cohort entry								
2008	0	<11[a]	<11[a]	NA	0	0	0	NA
2009	0	>369[a]	>369[a]	0.34	0	28 (2.5)	28 (1.3)	0.23
2010	29 (2.6)	441 (6.6)	470 (6.0)	0.19	29 (2.6)	48 (4.3)	77 (3.5)	0.09
2011	61 (5.4)	487 (7.3)	548 (7.0)	0.08	61 (5.5)	39 (3.5)	100 (4.5)	0.10
2012	73 (6.5)	482 (7.2)	555 (7.1)	0.03	73 (6.6)	54 (4.9)	127 (5.7)	0.07
2013	89 (7.9)	473 (7.1)	562 (7.2)	0.03	89 (8.0)	74 (6.6)	163 (7.3)	0.05
2014	95 (8.5)	481 (7.2)	576 (7.4)	0.05	95 (8.5)	75 (6.7)	170 (7.6)	0.07
2015	104 (9.3)	579 (8.7)	683 (8.8)	0.02	104 (9.3)	93 (8.4)	197 (8.8)	0.03
2016	150 (13.4)	742 (11.1)	892 (11.4)	0.07	150 (13.5)	134 (12.0)	284 (12.8)	0.04
2017	198 (17.7)	1027 (15.4)	1225 (15.7)	0.06	197 (17.7)	226 (20.3)	423 (19.0)	0.07
2018	225 (20.1)	1156 (17.3)	1381 (17.7)	0.07	222 (19.9)	235 (21.1)	457 (20.5)	0.03
2019	96 (8.6)	432 (6.5)	528 (6.8)	0.08	93 (8.4)	107 (9.6)	200 (9.0)	0.04
State								
Alabama	34 (3.0)	170 (2.5)	204 (2.6)	0.03	34 (3.1)	31 (2.8)	65 (2.9)	0.02
Arizona	21 (1.9)	107 (1.6)	128 (1.6)	0.02	20 (1.8)	21 (1.9)	41 (1.8)	0.01
California	12 (1.1)	118 (1.8)	130 (1.7)	0.06	12 (1.1)	21 (1.9)	33 (1.5)	0.07
Connecticut	24 (2.1)	138 (2.1)	162 (2.1)	0.01	23 (2.1)	24 (2.2)	47 (2.1)	0.01
Florida	164 (14.6)	1056 (15.8)	1220 (15.6)	0.03	164 (14.7)	157 (14.1)	321 (14.4)	0.02
Georgia	54 (4.8)	495 (7.4)	549 (7.0)	0.11	54 (4.9)	97 (8.7)	151 (6.8)	0.15
Illinois	62 (5.5)	318 (4.8)	380 (4.9)	0.04	62 (5.6)	58 (5.2)	120 (5.4)	0.02
Indian	39 (3.5)	177 (2.6)	216 (2.8)	0.05	39 (3.5)	34 (3.1)	73 (3.3)	0.03
Massachusetts	<11[a]	>90[a]	98 (1.3)	0.11	<11[a]	>11[a]	>11[a]	0.10
Minnesota	12 (1.1)	157 (2.4)	169 (2.2)	0.10	12 (1.1)	29 (2.6)	41 (1.8)	0.11
Missouri	50 (4.5)	275 (4.1)	325 (4.2)	0.02	50 (4.5)	44 (4.0)	94 (4.2)	0.03
North Carolina	>118[a]	>365[a]	498 (6.4)	0.20	124 (11.1)	54 (4.9)	178 (8.0)	0.23
New Jersey	75 (6.7)	234 (3.5)	309 (4.0)	0.15	74 (6.6)	45 (4.0)	119 (5.3)	0.12
New York	89 (7.9)	402 (6.0)	491 (6.3)	0.08	89 (8.0)	92 (8.3)	181 (8.1)	0.01
Ohio	33 (2.9)	331 (5.0)	364 (4.7)	0.10	33 (3.0)	47 (4.2)	80 (3.6)	0.07
Rhode Island	16 (1.4)	116 (1.7)	132 (1.7)	0.03	16 (1.4)	17 (1.5)	33 (1.5)	0.01
South Carolina	29 (2.6)	109 (1.6)	138 (1.8)	0.07	28 (2.5)	21 (1.9)	49 (2.2)	0.04
Tennessee	16 (1.4)	133 (2.0)	149 (1.9)	0.04	16 (1.4)	22 (2.0)	38 (1.7)	0.04
Texas	28 (2.5)	533 (8.0)	561 (7.2)	0.25	28 (2.5)	66 (5.9)	94 (4.2)	0.17
Utah	15 (1.3)	143 (2.1)	158 (2.0)	0.06	15 (1.3)	21 (1.9)	36 (1.6)	0.04
Virginia	<11[a]	>82[a]	>90[a]	0.07	<11[a]	>12[a]	>20[a]	0.08
Wisconsin	74 (6.6)	371 (5.6)	445 (5.7)	0.04	74 (6.6)	64 (5.8)	138 (6.2)	0.04
Other	137 (12.2)	744 (11.1)	881 (11.3)	0.03	135 (12.1)	118 (10.6)	253 (11.4)	0.05

Abbreviations: ACEi, angiotensin-converting-enzyme inhibitor; ARB, angiotensin II receptor blocker; CABG, coronary artery bypass grafting; COPD, chronic obstructive pulmonary disease; DOAC, direct-acting oral anticoagulant; ED, emergency department; eGFR, estimated glomerular filtration rate; IQR, interquartile range; MI, myocardial infarction; NA, not applicable; PAD, peripheral artery disease; PCI, percutaneous coronary intervention; PS, propensity score; PSA prostate-specific antigen; SMD, standardized mean difference.

SI conversion: To convert PSA level to micrograms per liter, multiply by 1.0.

[a] Cell suppression based on OptumLabs cell size suppression rules. Categories with fewer than 11 individuals are masked to protect patient confidentiality.

[b] Patients with PSA and without PSA levels were matched separately.

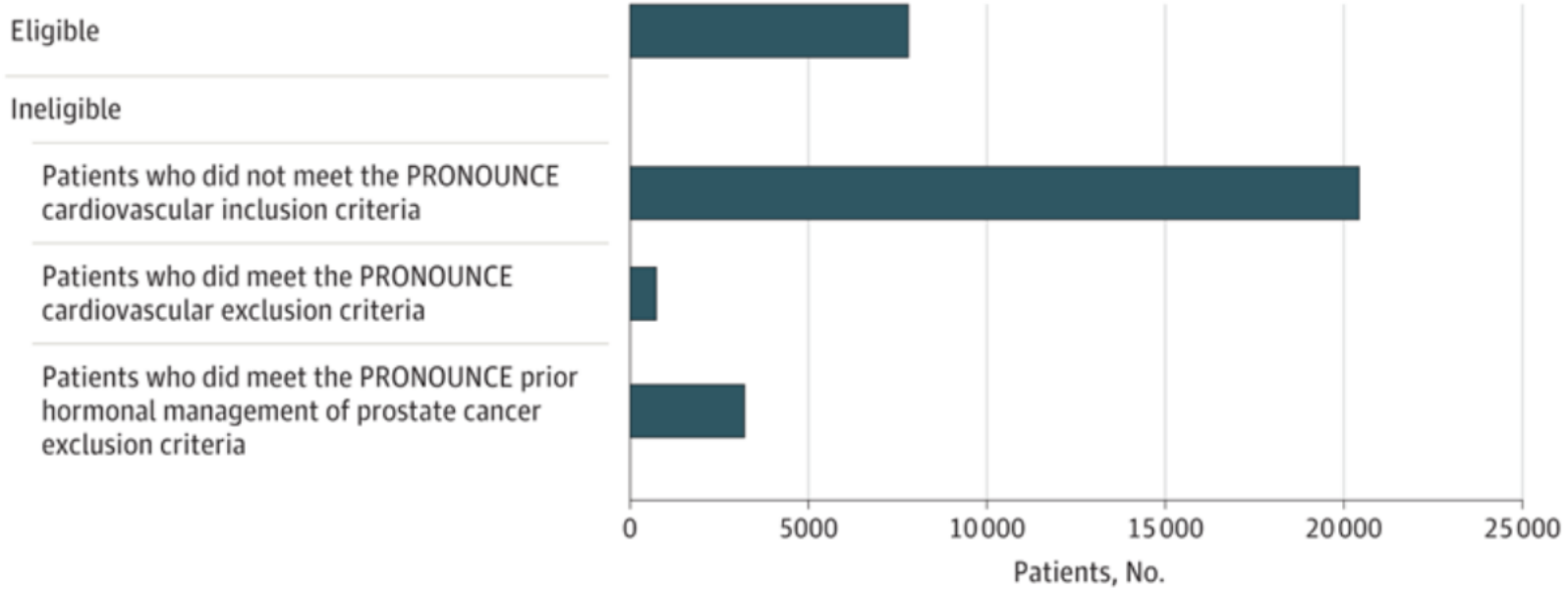
[c] eGFR was used for the subgroup analyses and not for matching purposes.

[d] Obesity defined as a body mass index (calculated as weight in kilograms divided by height in meters squared) of 30 or greater.

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Inclusion / Exclusion

Figure 1. Patients Included and Excluded After Applying Inclusion and Exclusion Criteria From the PRONOUNCE Trial (a Trial Comparing Cardiovascular Safety of Degarelix Versus Leuprolide in Patients With Advanced Prostate Cancer and Cardiovascular Disease)



[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Primary ITT Analyses

Table 2. End Points in 2226 Propensity-Matched Patients

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End point	Degarelix (n = 1113)			Leuprolide (n = 1113)			Hazard ratio (95% CI)	*P* value
	Events, No.	Person-years	Event rate per 100 person-years	Events, No.	Person-years	Event rate per 100 person-years		
Primary end point								
MACE[a]	88	864.65	10.18	73	849.15	8.60	1.18 (0.86-1.61)	.30
Secondary end points								
Death	65	875.05	7.43	43	861.81	4.99	1.48 (1.01-2.18)	.046
Myocardial infarction	19	869.63	2.18	16	855.18	1.87	1.16 (0.60-2.25)	.66
Stroke	15	869.75	1.72	16	855.78	1.87	0.92 (0.45-1.85)	.81
Angina	<11[b]	NA	NA	<11[b]	NA	NA	1.36 (0.43-4.27)	.60

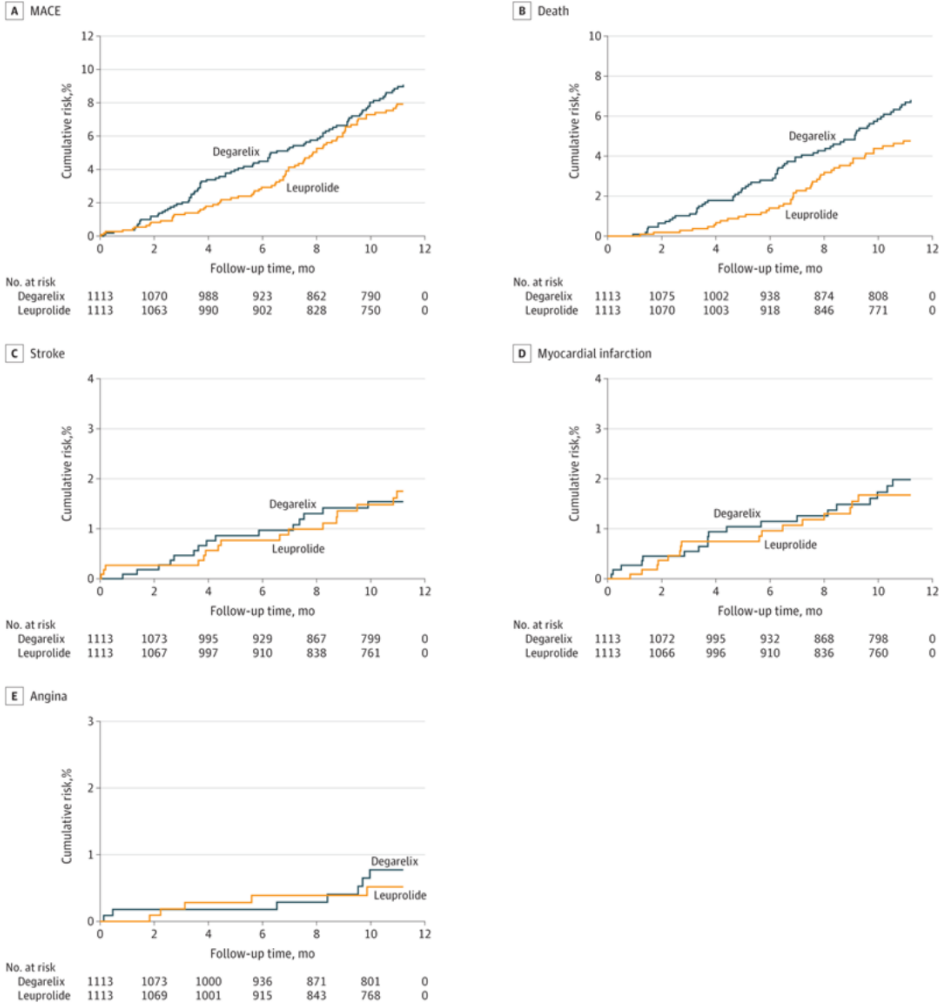
Abbreviations: MACE, major adverse cardiovascular event; NA, not applicable.

[a] MACE was defined as a composite end point of death from any cause, myocardial infarction, or stroke.

[b] Cell suppression based on OptumLabs cell size suppression rules. Categories with fewer than 11 individuals are masked to protect patient confidentiality.

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

Figure 2. Cumulative Incidence of Major Adverse Cardiovascular Event (MACE), Death, Stroke, Myocardial Infarction, and Angina in the Overall Cohort



TTE: Subgroup & Sensitivity Analysis

Table 3. End Points in Sensitivity Analyses

Table 3. End Points in Sensitivity Analyses

Outcome	Degarelix Patients, No.	Degarelix Events, No.	Degarelix Person-years	Degarelix Event rate, per 100 person-years	Leuprolide Patients, No.	Leuprolide Events, No.	Leuprolide Person-years	Leuprolide Event rate, per 100 person-years	Hazard ratio (95% CI)	*P* value
Patients who failed to meet the cardiovascular inclusion criteria										
MACE[a]	2952	112	2297.82	4.87	2952	92	2293.01	4.01	1.21 (0.92-1.60)	.17
Death	2952	83	2310.67	3.59	2952	54	2308.80	2.34	1.53 (1.09-2.16)	.02
Myocardial infarction	2952	12	2307.05	0.52	2952	23	2301.27	1.00	0.52 (0.26-1.04)	.06
Stroke	2952	27	2301.10	1.17	2952	27	2300.51	1.17	0.99 (0.58-1.70)	.98
Angina	2952	<11[b]	NA	NA	2952	<11[b]	NA	NA	0.66 (0.19-2.34)	.52
Patients who met the cardiovascular exclusion criteria										
MACE[a]	96	<11[b]	NA	NA	96	12	NA	NA	0.64 (0.27-1.56)	.33
Death	96	<11[b]	NA	NA	96	<11[b]	NA	NA	0.83 (0.29-2.44)	.74
Myocardial infarction	96	<11[b]	NA	NA	96	<11[b]	NA	NA	3.02 (0.32-28.87)	.34
Stroke	96	<11[b]	NA	NA	96	<11[b]	NA	NA	0.39 (0.08-2.00)	.26
Angina	96	<11[b]	NA	NA	96	<11[b]	NA	NA	NA	NA
Patients who crossed over between 2 treatments censored when they switched										
MACE[a]	1113	52	431.30	12.06	1113	72	847.11	8.50	1.58 (1.11-2.26)	.01
Death	1113	35	436.55	8.02	1113	42	859.76	4.89	1.95 (1.26-3.03)	.003
Myocardial infarction	1113	13	433.96	3.00	1113	16	853.14	1.88	1.57 (0.74-3.35)	.24
Stroke	1113	<11[b]	NA	NA	1113	16	NA	NA	1.10 (0.48-2.53)	.81
Angina	1113	<11[b]	NA	NA	1113	<11[b]	NA	NA	1.45 (0.39-5.43)	.58
Patients who crossed over between treatments excluded										
MACE[a]	398	47	289.85	16.22	398	32	310.44	10.31	1.58 (1.01-2.47)	.046
Death	398	35	294.34	11.89	398	17	314.92	5.40	2.21 (1.24-3.94)	.01
Myocardial infarction	398	<11[b]	NA	NA	398	14	310.58	4.51	0.66 (0.28-1.52)	.33
Stroke	398	<11[b]	NA	NA	398	<11[b]	NA	NA	1.67 (0.55-5.11)	.37
Angina	398	<11[b]	NA	NA	398	<11[b]	NA	NA	0.51 (0.09-2.79)	.44
IPTW instead of propensity-score matching										
MACE[a]	1120	593	6194.59	9.58	6680	531	6135.66	8.65	1.10 (0.86-1.41)	.43
Death	1120	460	6252.48	7.36	6680	304	6233.80	4.87	1.50 (1.13-2.01)	.01
Myocardial infarction	1120	132	6218.51	2.13	6680	139	6180.94	2.24	0.94 (0.56-1.58)	.82
Stroke	1120	89	6225.89	1.43	6680	128	6187.43	2.06	0.69 (0.38-1.24)	.21
Angina	1120	44	6235.09	0.71	6680	42	6214.76	0.68	1.03 (0.44-2.42)	.95
PDC ≥80%										
MACE[a]	690	50	523.12	9.56	690	52	494.69	10.51	0.90 (0.61-1.33)	.61
Death	690	36	530.74	6.78	690	31	505.73	6.13	1.09 (0.68-1.77)	.71
Myocardial infarction	690	13	526.60	2.47	690	15	498.44	3.01	0.83 (0.39-1.74)	.62
Stroke	690	<11[b]	NA	NA	690	<11[b]	NA	NA	1.07 (0.41-2.78)	.89
Angina	690	<11[b]	NA	NA	690	<11[b]	NA	NA	2.84 (0.58-13.99)	.20
PDC <80%										
MACE[a]	417	38	336.90	11.28	417	25	352.12	7.10	1.60 (0.97-2.64)	.07
Death	417	29	339.69	8.54	417	19	353.18	5.38	1.60 (0.90-2.85)	.11
Myocardial infarction	417	<11[b]	NA	NA	417	<11[b]	NA	NA	3.10 (0.63-15.60)	.16
Stroke	417	<11[b]	NA	NA	417	<11[b]	NA	NA	1.02 (0.33-3.17)	.97
Angina	417	<11[b]	NA	NA	417	<11[b]	NA	NA	0.34 (0.04-3.32)	.36

Abbreviations: IPTW, inverse probability of treatment weighting; MACE, major adverse cardiovascular event; NA, not applicable; PDC, proportion of days covered.

[a] MACE was defined as a composite end point of death from any cause, myocardial infarction, or stroke.

[b] Cell suppression based on OptumLabs cell size suppression rules. Categories with fewer than 11 individuals are masked to protect patient confidentiality.

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: Conclusions

- In this emulated clinical trial of men with commercial insurance or Medicare Advantage in the United States who had cardiovascular disease and initiated Androgen deprivation therapy (ADT) for prostate cancer, **degarelix was not associated with a lower risk of cardiovascular events vs leuprolide**, **in contrast to prior reports**
- We observed **frequent crossover between medications** and **missing data on disease characteristics** in our cohort, which may have contributed to residual confounding and highlights some of the challenges of using RWD to emulate RCTs
- The population included in this study as well as the findings should be compared with those of the PRONOUNCE trial, once available, to assess the fidelity of results from this real-world observational study with reported trial end points and to deepen our understanding of the optimal role of using RWD to emulate RCTs

[Reference] Wallach JD, Deng Y, McCoy RG, et al. Real-world Cardiovascular Outcomes Associated With Degarelix vs Leuprolide for Prostate Cancer Treatment. *JAMA Netw Open*. 2021;4(10):e2130587. Published 2021 Oct 1. doi:10.1001/jamanetworkopen.2021.30587

TTE: RCT-DUPLICATE Initiative

- Emulation of Randomized Clinical Trials With Nonrandomized Database Analyses: Results of 32 Clinical Trials
  - JAMA: 2023-04-25



Research

JAMA | Original Investigation

Emulation of Randomized Clinical Trials With Nonrandomized Database Analyses
Results of 32 Clinical Trials

Shirley V. Wang, PhD, ScM; Sebastian Schneeweiss, MD, ScD; and the RCT-DUPLICATE Initiative

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Supplemental content

IMPORTANCE Nonrandomized studies using insurance claims databases can be analyzed to produce real-world evidence on the effectiveness of medical products. Given the lack of baseline randomization and measurement issues, concerns exist about whether such studies produce unbiased treatment effect estimates.

OBJECTIVE To emulate the design of 30 completed and 2 ongoing randomized clinical trials (RCTs) of medications with database studies using observational analogues of the RCT design parameters (population, intervention, comparator, outcome, time [PICOT]) and to quantify agreement in RCT-database study pairs.

DESIGN, SETTING, AND PARTICIPANTS New-user cohort studies with propensity score matching using 3 US claims databases (Optum Clinformatics, MarketScan, and Medicare). Inclusion-exclusion criteria for each database study were prespecified to emulate the corresponding RCT. RCTs were explicitly selected based on feasibility, including power, key confounders, and end points more likely to be emulated with real-world data. All 32 protocols were registered on ClinicalTrials.gov before conducting analyses. Emulations were conducted from 2017 through 2022.

EXPOSURES Therapies for multiple clinical conditions were included.

MAIN OUTCOMES AND MEASURES Database study emulations focused on the primary outcome of the corresponding RCT. Findings of database studies were compared with RCTs using predefined metrics, including Pearson correlation coefficients and binary metrics based on statistical significance agreement, estimate agreement, and standardized difference.

RESULTS In these highly selected RCTs, the overall observed agreement between the RCT and the database emulation results was a Pearson correlation of 0.82 (95% CI, 0.64-0.91), with 72% meeting statistical significance, 66% estimate agreement, and 75% standardized difference agreement. In a post hoc analysis limited to 16 RCTs with closer emulation of trial design and measurements, concordance was higher (Pearson *r*, 0.93; 95% CI, 0.79-0.97; 94% meeting statistical significance, 88% estimate agreement, 88% standardized difference agreement). Weaker concordance occurred among 16 RCTs for which close emulation of certain design elements that define the research question (PICOT) with data from insurance claims was not possible (Pearson *r*, 0.53; 95% CI, 0.00-0.83; 50% meeting statistical significance, 50% estimate agreement, 69% standardized difference agreement).

CONCLUSIONS AND RELEVANCE Real-world evidence studies can reach similar conclusions as RCTs when design and measurements can be closely emulated, but this may be difficult to achieve. Concordance in results varied depending on the agreement metric. Emulation differences, chance, and residual confounding can contribute to divergence in results and are difficult to disentangle.

Author Affiliations: Division of Pharmacoepidemiology and

[Reference] Wang SV, Schneeweiss S; RCT-DUPLICATE Initiative, et al. Emulation of Randomized Clinical Trials With Nonrandomized Database Analyses: Results of 32 Clinical Trials. *JAMA*. 2023;329(16):1376-1385. doi:10.1001/jama.2023.4221

TTE: RCT-DUPLICATE Initiative (32 Studies)

Table 1. Effect Estimates and Agreement Metrics

Table 1. Effect Estimates and Agreement Metrics

Study No.	Trial name	Effect estimates (95% CI)			Standardized difference[c]	Agreement		
		RCT	Database studyAdjusted[a,b]	Database study crude[a,b]		Statistical significance	Estimate	Standardized difference
1	LEADER	0.87 (0.78 to 0.97)	0.82 (0.76 to 0.87)	0.57 (0.54 to 0.61)	0.90	SA	EA	SD
2	DECLARE-TIMI58	0.83 (0.73 to 0.95)	0.69 (0.59 to 0.81)	0.47 (0.41 to 0.53)	1.76	SA		SD
3	EMPA-REG	0.86 (0.74 to 0.99)	0.83 (0.73 to 0.95)	0.63 (0.57 to 0.70)	0.35	SA	EA	SD
4	CANVAS	0.86 (0.75 to 0.97)	0.77 (0.70 to 0.85)	0.58 (0.54 to 0.62)	1.34	SA	EA	SD
5	CARMELINA	1.02 (0.89 to 1.17)	0.90 (0.84 to 0.96)	0.90 (0.86 to 0.95)	1.61	SAP	EA	SD
6	TECOS	0.98 (0.88 to 1.09)	0.89 (0.86 to 0.91)	0.81 (0.79 to 0.84)	1.71	SAP	EA	SD
7	SAVOR-TIMI	1.00 (0.89 to 1.12)	0.81 (0.76 to 0.86)	0.65 (0.62 to 0.69)	3.16	SAP		
8	LEAD-2	0 (−0.20 to 0.20)	0.05 (−0.11 to 0.22)	0.01 (−0.11 to 0.13)	−0.37	SA	EA	SD
9	TRITON-TIMI	0.81 (0.73 to 0.90)	0.88 (0.79 to 0.97)	0.70 (0.65 to 0.76)	−1.11	SA	EA	SD
10	PLATO	0.84 (0.77 to 0.92)	0.92 (0.83 to 1.02)	0.84 (0.78 to 0.91)	−1.31		EA	SD
11	ISAR-REACT 5	1.36 (1.09 to 1.70)	NA[d]	NA[d]	NA[d]	NA[d]		
12	ARISTOTLE	0.79 (0.66 to 0.95)	0.68 (0.61 to 0.76)	0.66 (0.62 to 0.71)	1.36	SA	EA	SD
13	RE-LY	0.66 (0.53 to 0.82)	0.73 (0.60 to 0.90)	0.67 (0.58 to 0.78)	−0.66	SA	EA	SD
14	ROCKET AF	0.79 (0.66 to 0.96)	0.70 (0.62 to 0.80)	0.76 (0.69 to 0.84)	1.00	SA	EA	SD
15	EINSTEIN DVT	0.68 (0.44 to 1.04)	0.75 (0.62 to 0.90)	0.85 (0.76 to 0.95)	−0.42	SAP	EA	SD
16	EINSTEIN PE	1.12 (0.75 to 1.68)	0.67 (0.55 to 0.80)	0.73 (0.64 to 0.83)	2.28	SAP		
17	RE-COVER II	1.08 (0.64 to 1.80)	1.15 (0.74 to 1.78)	1.48 (1.09 to 2.00)	−0.18	SA	EA	SD
18	AMPLIFY	0.84 (0.60 to 1.18)	0.81 (0.54 to 1.23)	0.64 (0.50 to 0.82)	0.13	SA	EA	SD
19	RECORD1	0.25 (0.14 to 0.47)	0.17 (0.10 to 0.29)	0.25 (0.18 to 0.34)	0.63	SA	EA	SD
20	TRANSCEND	0.92 (0.81 to 1.05)	0.88 (0.81 to 0.96)	0.80 (0.74 to 0.85)	0.55		EA	SD
21	ONTARGET	1.01 (0.94 to 1.09)	0.83 (0.77 to 0.90)	0.68 (0.64 to 0.72)	3.46	SAP		
22	HORIZON-PFT	0.59 (0.42 to 0.83)	0.72 (0.55 to 0.94)	1.08 (0.86 to 1.35)	−0.90	SA	EA	SD
23	VERO	0.44 (0.29 to 0.68)	NA[d]	NA[d]	NA[d]	NA[d]	NA[d]	NA[d]
24	DAPA-CKD	0.61 (0.51 to 0.72)	0.80 (0.52 to 1.26)	0.41 (0.29 to 0.58)	−1.10	SD		SD
25	PARADIGM-HF	0.80 (0.73 to 0.87)	1.02 (0.91 to 1.14)	0.95 (0.90 to 1.02)	−3.42			
26	P04334[e,f]	0.56 (0.44 to 0.72)	0.78 (0.62 to 0.97)	0.87 (0.76 to 0.99)	−1.95	SA	SD	SD
27	D5896	1.07 (0.70 to 1.65)	1.38 (0.90 to 2.13)	1.41 (1.00 to 1.98)	−0.81	SA	EA	SD
28	IMPACT[e,g]	0.85 (0.80 to 0.90)	1.13 (1.04 to 1.23)	1.22 (1.15 to 1.30)	−5.46			
29	POET-COPD	0.83 (0.77 to 0.90)	1.02 (0.93 to 1.12)	1.05 (0.99 to 1.12)	−3.27			
30	INSPIRE[h]	0.97 (0.84 to 1.12)	0.93 (0.90 to 0.96)	0.83 (0.81 to 0.85)	0.56	SA	EA	SD
31	CAROLINA[i]	0.98 (0.84 to 1.14)	0.91 (0.79 to 1.05)	0.92 (0.83-1.01)	0.70	SA	EA	SD
32	PRONOUNCE[i]	1.28 (0.59 to 2.79)	1.35 (0.94 to 1.93)	1.70 (1.30 to 2.21)	−0.12	SA	EA	SD

Abbreviations: EA, estimate agreement, adjusted database study point estimates falling within the 95% CI of the corresponding randomized clinical trial (RCT) result; NA, not applicable; SA, full statistical significance agreement, adjusted database study and RCT estimates and CIs on the same side of null; SAP, partial significance agreement, meets the prespecified noninferiority criteria even though the database study may have indicated superiority; SD, standardized difference agreement, SDs $|z|$ less than 1.96.

[a] Pooled across databases.

[b] Crude estimates for the database study cohorts designed to emulate trial designs do not adjust for confounding except through design. Adjusted estimates additionally adjust for confounding through propensity score matching on prespecified risk factors for the outcome that are associated with exposure.

[c] The SD calculations are available in the Methods section. These quantify the difference in effect size between the RCT and database study relative to the pooled standard deviation. Therefore, an SD of 1.00 indicates that the effect estimate from the RCT and the database study are 1 standard deviation apart. Assuming an α level of .05 and assuming that both the database and RCT results are based on large samples, the null hypothesis of no difference would be rejected whenever $|z|>1.96$.

[d] χ^2 test indicated that results were heterogeneous across databases. See database-specific results in Supplement 1.

[e] Trial had coprimary comparisons. The first listed was the primary comparison in the database study emulation protocol.

[f] Because an effect estimate was not reported, we calculated the risk ratio based on results reported for P4334 (PMID: 20678306).

[g] Because of challenges with measurement of recurrent outcomes, the estimated hazard ratio from secondary analyses of time to first occurrence of the primary outcome was used as the comparison for the database study emulation of the IMPACT trial instead of the rate ratio.

[h] Because of challenges with measurement of recurrent outcomes and low recurrence rate, the estimated rate ratio from the INSPIRE trial was approximated with a hazard ratio from the database emulation of the RCT.

[i] PRONOUNCE and CAROLINA were the 2 trials for which the trial design was emulated, protocol registered, and database study results generated before the results of the trials were made public.

[Reference] Wang SV, Schneeweiss S; RCT-DUPLICATE Initiative, et al. Emulation of Randomized Clinical Trials With Nonrandomized Database Analyses: Results of 32 Clinical Trials. *JAMA*. 2023;329(16):1376-1385. doi:10.1001/jama.2023.4221

TTE: Emulation Challenges

No.	Trial name	Comparator emulation[a]	Outcome emulation[b]	Age distribution, mean difference, y	Sex distribution, difference in % female	Run-in window[c]	Placebo control	In-hospital start of medication	Dose titration during follow-up	Discontinuation of maintenance therapy at randomization	Delayed effect[d]	Close emulation[e]
1	LEADER	Moderate	Good	-3.4	-17.8	Yes, placebo	Yes	No	Yes	No	No	Yes
2	DECLARE	Moderate	Moderate	1.4	-4.9	Yes, placebo	Yes	No	No	No	No	No
3	EMPA-REG	Moderate	Good	1.2	-11.9	Yes, placebo	Yes	No	No	No	No	Yes
4	CANVAS	Moderate	Good	-2.0	-10.5	Yes, placebo	Yes	No	No	No	No	Yes
5	CARMELINA	Poor	Good	-6.4	-16.2	No	Yes	No	No	No	No	No
6	TECOS	Poor	Moderate	-6.8	-18.1	No	Yes	No	No	No	No	No
7	SAVOR-TIMI	Poor	Good	-3.8	-13.7	No	Yes	No	No	No	No	No
8	LEAD-2	Good	Moderate	-2.0	-6.0	Yes, both groups	No	No	Yes	No	No	Yes
9	TRITON-TIMI 38	Good	Good	3.4[f]	4.9	No	No	Yes	Yes	No	No	No
10	PLATO	Good	Good	-3.3[f]	-4.1	No	No	Yes	Yes	No	No	No
11	ISAR-REACT 5	Good	Good	5.6	0.9	No	No	Yes	Yes	No	No	No
12	ARISTOTLE	Good	Good	-6.1	-16.7	No	No	No	No	Yes[g]	No	Yes
13	RE-LY	Good	Good	-4.7	-5.9	No	No	No	No	Yes[g]	No	Yes
14	ROCKET-AF	Good	Good	-4.5	-14.9	No	No	No	No	Yes[g]	No	Yes
15	EINSTEIN DVT	Good	Moderate	-14.7	-17.0	No	No	No	Yes	No	No	Yes
16	EINSTEIN PE	Good	Moderate	-8.2	-4.9	No	No	No	Yes	No	No	Yes
17	RE-COVER II	Good	Moderate	-13.5	-16.4	No	No	No	No	No	No	Yes
18	AMPLIFY	Good	Moderate	-0.6	-10.1	No	No	No	Yes	No	No	Yes
19	RECORD1	Good	Good	1.0	1.6	No	No	No	No	No	No	Yes
20	TRANSCEND	Moderate	Good	-4.0	-14.1	Yes, both groups	Yes	No	No	No	No	Yes
21	ON TARGET	Good	Good	-2.4	-27.2	Yes, both groups	No	No	Yes	No	No	Yes
22	HORIZON PFT	Moderate	Good	-1.0	0	No	Yes	No	No	No	Yes	No
23	VERO	Good	Moderate	1.1	0	No	No	No	No	No	Yes	No
24	DAPA-CKD	Moderate	Moderate	-5.5	-11.4	No	Yes	No	No	No	Yes	No
25	PARADIGM-HF	Moderate	Moderate	-4.7	-6.2	Yes, both groups	No	No	No	Yes	No	No
26	P04334	Good	Good	-11.2	1.9	Yes, 1 class	No	No	No	Yes	No	No
27	D5896	Good	Good	-3.3	-1.8	No	No	No	No	Yes	No	No
28	IMPACT	Good	Good	-4.0	-25.5	Yes, baseline prescription	No	No	No	Yes	No	No
29	POET-COPD	Good	Good	-7.5	-28.3	Yes, mixed	No	No	No	Yes	No	No
30	INSPIRE[h]	Good	Moderate	-1.5	-44.6	Yes, 1 class	No	No	No	Yes	No	No
31	CAROLINA[i]	Good	Good	-6.3	-12.3	Yes, placebo	No	No	Yes	No	No	Yes
32	PRONOUNCE1[i]	Good	Good	-3.0	0	No	No	No	Yes	No	No	Yes

[Reference] Wang SV, Schneeweiss S; RCT-DUPLICATE Initiative, et al. Emulation of Randomized Clinical Trials With Nonrandomized Database Analyses: Results of 32 Clinical Trials. *JAMA*. 2023;329(16):1376-1385. doi:10.1001/jama.2023.4221

TTE: Limitations → Further Considerations

1. We assumed that the findings from a **single RCT were internally valid**, which is not guaranteed.
2. Apparent agreement between RCT and database study results could occur if the effects of multiple factors (**chance, emulation differences, bias**) **cancel each other out**
3. The results of 32 database studies that emulate RCTs may have **limited generalizability** due to the multistep selection process and feasibility requirements
4. Although our goal with this project was to calibrate database study results against RCTs, in practice, **many highly controlled trials cannot be emulated with database studies** and **many questions of interest may never have trials to calibrate against**

→ The principles applied in this project remain fundamental for the interpretation of database study results; namely, **to specify a hypothetical trial** that would answer the study question and **to assess robustness through thoughtful sensitivity analyses**

[Reference] Wang SV, Schneeweiss S; RCT-DUPLICATE Initiative, et al. Emulation of Randomized Clinical Trials With Nonrandomized Database Analyses: Results of 32 Clinical Trials. *JAMA*. 2023;329(16):1376-1385. doi:10.1001/jama.2023.4221


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